

Assignment 5

Loops In JAVA

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Objective

The objective of this set of programs is to practice fundamental Java concepts such as loops (for, while), conditional statements (if-else), and basic arithmetic operations by solving simple number-based problems.

1. Find the Sum of Numbers from 1 to N

This program reads a value N from the user and calculates the sum of all natural numbers from 1 to N using a for loop.

Java Program

```
import java.util.Scanner;

public class SumOfNumbers {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the value of N: ");
        int n = sc.nextInt();

        int sum = 0;
        for (int i = 1; i <= n; i++) {
            sum = sum + i;
        }

        System.out.println("Sum of numbers from 1 to " + n + " = " + sum);

        sc.close();
    }
}
```

Output

```
Enter the value of N: 10
Sum of numbers from 1 to 10 = 55
```

2. Print Multiples of 5 from 1 to 100

This program iterates through numbers from 1 to 100 and prints only those numbers that are exactly divisible by 5.

Java Program

```
public class MultiplesOfFive {  
    public static void main(String[] args) {  
        System.out.println("Multiples of 5 from 1 to 100:");  
  
        for (int i = 1; i <= 100; i++) {  
            if (i % 5 == 0) {  
                System.out.print(i + " ");  
            }  
        }  
        System.out.println();  
    }  
}
```

Output

```
Multiples of 5 from 1 to 100:  
5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100
```

3. Count How Many Numbers Are Divisible by 3 Between 1 and N

This program reads a value N from the user and counts how many numbers between 1 and N are exactly divisible by 3.

Java Program

```
import java.util.Scanner;

public class CountDivisibleByThree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the value of N: ");
        int n = sc.nextInt();

        int count = 0;
        for (int i = 1; i <= n; i++) {
            if (i % 3 == 0) {
                count++;
            }
        }

        System.out.println("Count of numbers divisible by 3 between 1 and " + n + " = " + count);

        sc.close();
    }
}
```

Output

```
Enter the value of N: 20
Count of numbers divisible by 3 between 1 and 20 = 6
```

4. Check Whether a Number Is Prime or Not

This program reads a number from the user and checks whether it is a prime number by testing divisibility from 2 up to the square root of the number.

Java Program

```
import java.util.Scanner;

public class PrimeCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        boolean isPrime = true;

        if (num <= 1) {
            isPrime = false;
        } else {
            for (int i = 2; i <= Math.sqrt(num); i++) {
                if (num % i == 0) {
                    isPrime = false;
                    break;
                }
            }
        }

        if (isPrime) {
            System.out.println(num + " is a Prime Number");
        } else {
            System.out.println(num + " is Not a Prime Number");
        }

        sc.close();
    }
}
```

Output

```
Enter a number: 29
29 is a Prime Number

$ java PrimeCheck.java
Enter a number: 15
15 is Not a Prime Number
```

5. Reverse of a Given Number

This program reads a number from the user and reverses its digits using the modulus and division operators inside a while loop.

Java Program

```
import java.util.Scanner;

public class ReverseNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        int original = num;
        int reversed = 0;

        while (num != 0) {
            int digit = num % 10;
            reversed = reversed * 10 + digit;
            num = num / 10;
        }

        System.out.println("Reverse of " + original + " is " + reversed);

        sc.close();
    }
}
```

Output

```
Enter a number: 12345
Reverse of 12345 is 54321
```